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Introduction

This report is based on my practical work that I completed during the Operating Systems and Networks module using Cisco Packet Tracer and an Ubuntu Linux VirtualBox Machine. The aim of this report is to explain the main networking and operating system concepts that were covered throughout the module. These technologies are used in real workplaces to keep networks running and secure.

The report is split into four sections. Section 1 explains the network design and routing configuration from the practical test. Section 2 explains Linux fundamentals including the directory structure, commands, and file permissions used. Section 3 looks at how DHCP works and how it was configured in Packet Tracer. Section 4 explores VLANs and how they are used to segment networks. Detailed screenshots from the practical lab work are included throughout the report as evidence.

Section 1: Network Design and Routing

1.1 What is a Routed Network and Why is Routing Needed

Throughout my learning I understood that a routed network is a network where multiple different IP networks are connected through routers. In my practical test topology, two LANs were connected through two routers (Router0 and Router1), with switches connecting the end devices on each side. Without routing, devices on one network cannot communicate with devices on a different network at Layer 3. The router acts as a gateway between the two networks, receiving packets and forwarding them to the correct destination. Devices can only communicate directly on the same network.

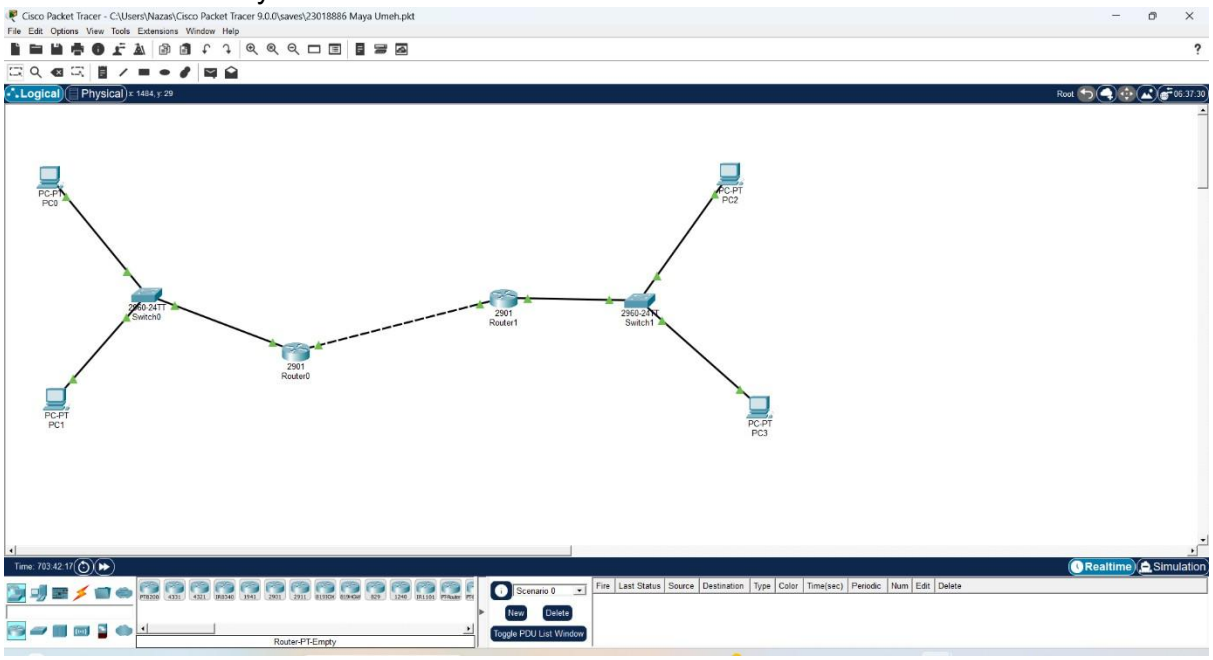


Figure 1: Network topology showing routers, switches and PCs

1.2 How IPv4 Addressing Works

Every device in the network needs its own IPv4 address so it can send and receive data properly. The IP address is split into two parts: the network part and the host part. The network part shows which network the device is in, and the host part identifies the actual

device. The subnet mask is what separates these two, for example 255.255.255.0 where the last part is used for hosts.

In my network, I used addresses from the 192.168.86.0 range. When I configured the routers, I had to assign IP addresses to each interface so they could connect to different networks.

```
Router>
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.86.1    YES manual up                    up
GigabitEthernet0/1     192.168.86.129  YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

Figure 2: Shows the output of *show ip interface brief* showing configured IP addresses

The screenshot shows that, Router0 has two interfaces with different IP addresses. GigabitEthernet0/0 was set to 192.168.86.1 and GigabitEthernet0/1 was set to 192.168.86.129. From this, I could see that each interface is connected to a different network.

This is important because the router needs one IP per network to be able to route traffic between them. Without assigning these addresses correctly, the router would not know how to forward packets from one network to another.

1.3 Why Devices in Different Networks Cannot Communicate Without Routing?

From my practical, this confirmed that devices in different networks cannot communicate directly because they are not in the same IP range. When I tried sending data between devices on different networks, it did not work unless a router was involved.

When a device wants to send data, it first checks if the destination IP is in the same network as itself. If it is, it sends the data directly. But if it is not in the same network, the device sends the data to its default gateway, which is the router. The router then uses its routing table to determine the most appropriate path for forwarding the packet. Without this step, the device would not know where to send the data, and the communication would fail.

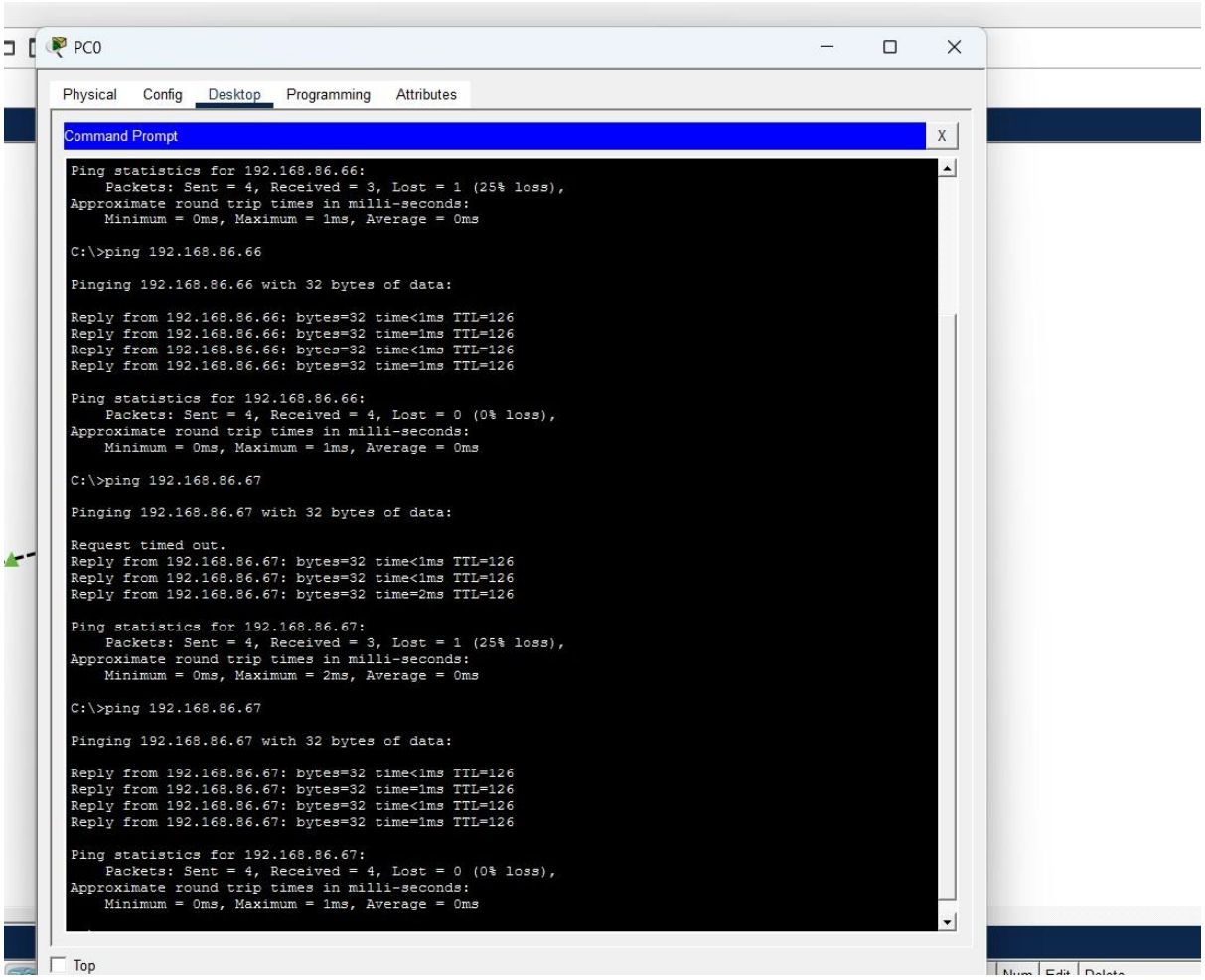


Figure 3: Ping test between devices on different networks

From my screenshot, I could see that communication between networks only worked when routing was correctly set up. This helped me understand that the router is essential because it connects separate networks together and allows them to communicate.

1.4 Static Routing vs Dynamic Routing

From what I understood in the practical, static routing is when you manually add routes into the router. This means you must type in every network yourself, which becomes inefficient for my topology as multiple networks are connected and routes will need to be updated manually if changes occur.

Dynamic routing is more suitable in my topology because routers share information automatically, reducing manual configuration as the network grows. Instead of adding everything manually, the routers update their routing tables by themselves.

RIP is one example of dynamic routing. It sends updates between routers so they can learn about other networks without needing manual configuration.

1.5 Why RIP Was Used in My Network

In my practical, I used RIP as the routing protocol.

```
Router>
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.86.0/24 is variably subnetted, 5 subnets, 3 masks
C       192.168.86.0/26 is directly connected, GigabitEthernet0/0
L       192.168.86.1/32 is directly connected, GigabitEthernet0/0
R       192.168.86.64/26 [120/1] via 192.168.86.130, 00:00:16, GigabitEthernet0/1
C       192.168.86.128/30 is directly connected, GigabitEthernet0/1
L       192.168.86.129/32 is directly connected, GigabitEthernet0/1

Router>ZS
```

Figure 4: Routing table showing RIP routes

From my screenshot, I could see an “R” in the routing table, which shows that the route was learned using RIP. For example, the network 192.168.86.64/26 was learned through another router, which means the routers were sharing information automatically.

I used RIP because it was simple to set up and worked well for my small network, although it is mainly suitable for smaller networks due to its slower convergence and limited scalability.

Section 2: Operating Systems Theory: Linux Fundamentals

2.1 The Linux Directory Structure

From my practical work using the Ubuntu virtual machine, I could see that Linux stores everything in one main structure starting from the root directory, which is shown as /. To explore this, I used the command ls / in the terminal to display all the main directories in the system.

```
cu23aan@23018886-VirtualBox:~$ ls /
bin                cdrom  home   lib.usr-is-merged  mnt   root  sbin.usr-is-merged  swap.img  usr
bin.usr-is-merged  dev    lib    lost+found         opt   run   snap                sys       var
boot               etc    lib64  media              proc  sbin  srv                 tmp
cu23aan@23018886-VirtualBox:~$ ls /home
cu23aan  testuser
cu23aan@23018886-VirtualBox:~$ ls /var
backups  cache  crash  lib  local  lock  log  mail  metrics  opt  run  snap  spool  tmp
cu23aan@23018886-VirtualBox:~$ ls /etc | head
adduser.conf
alsa
alternatives
anacrontab
apg.conf
apm
apparmor
apparmor.d
apport
apt
cu23aan@23018886-VirtualBox:~$ /bin
```

Figure 5: Output of ls / showing the root directory structure

The screenshot shows key directories including /home, /etc, /var and /bin as part of the root structure. Each of these has a specific role in how the Linux system operates, helping organise system files, user data, and essential commands efficiently.

The /home directory is where user files are stored. For example, my own files are saved inside my user folder in this location. The /etc directory contains configuration files that control how the system works. The /var directory stores data that changes often, such as log files. The /bin directory contains essential commands that are used in the terminal, such as basic system commands.

This helped me understand that Linux is organised in a structured way, where each directory has a specific role within the overall system structure.

2.2 Files and Directories in Linux

From using the terminal, I learned that everything in Linux is either a file or a directory. To see the difference, I used the command ls -l, which shows more detailed information about each item.

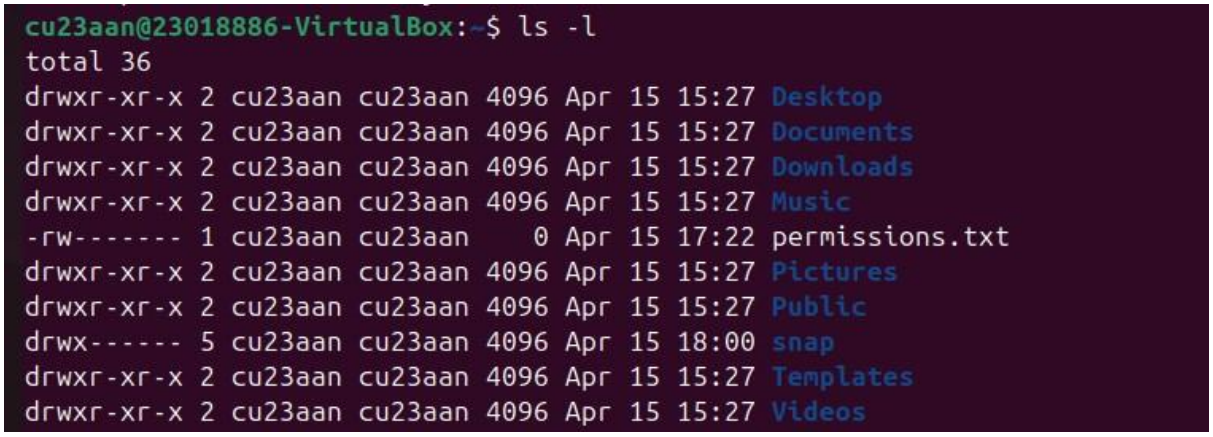


Figure 6: Output of ls -l showing files and directories

From my screenshot, I could see entries starting with d such as drwxr-xr-x, which shows that those are directories. The “d” at the beginning stands for directory, while files would start with a “-” instead. The rest of the letters represent permissions, but the first character is what tells you the type.

Directories are used to store and organise files, like folders, while files contain the actual data such as text, programs, or configuration settings. Understanding this was important because directories and files are affected differently by permissions, especially when controlling access and execution.

2.3 Why the Command Line is Important?

The command line is the main way to interact with the system. Instead of clicking through menus like in Windows, everything is done by typing commands in the terminal.

During the practical sessions, I used commands like ls, whoami and chmod to move around the system and check information. This made it much quicker to navigate compared to using a graphical interface. Many real servers have no graphical interface at all, which is why the command line is essential. Another thing I noticed is that the command line gives more control. For example, instead of opening folders manually, I could directly go to a location using one command.

It also allows commands to be repeated and automated, which saves time when doing tasks multiple times. Overall, the command line is important because it is faster, more efficient, and gives full control over the system.

2.4 Users, Groups and Permissions

Linux is a multi-user operating system, which means multiple users can exist on the same system. During the lab, I used the terminal to check and manage users.

First, I used the command: whoami

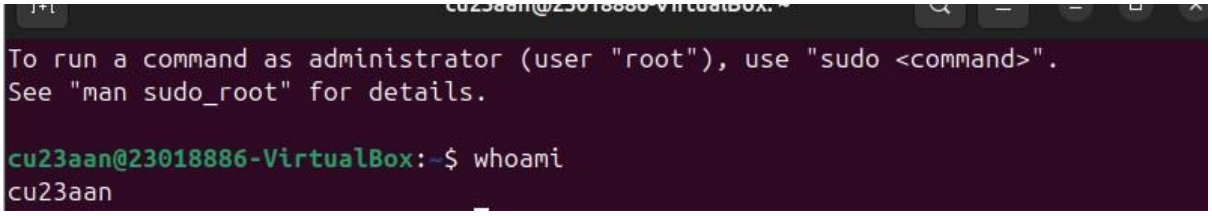


Figure 7: Output of whoami showing the currently logged-in user(cu23aan)

From my screenshot, I could see that the username cu23aan was returned, which confirmed that this was the active user I was working under.

After that, I created a new user using: sudo adduser testuser

Then I checked the groups that the new user belongs to using groups testuser

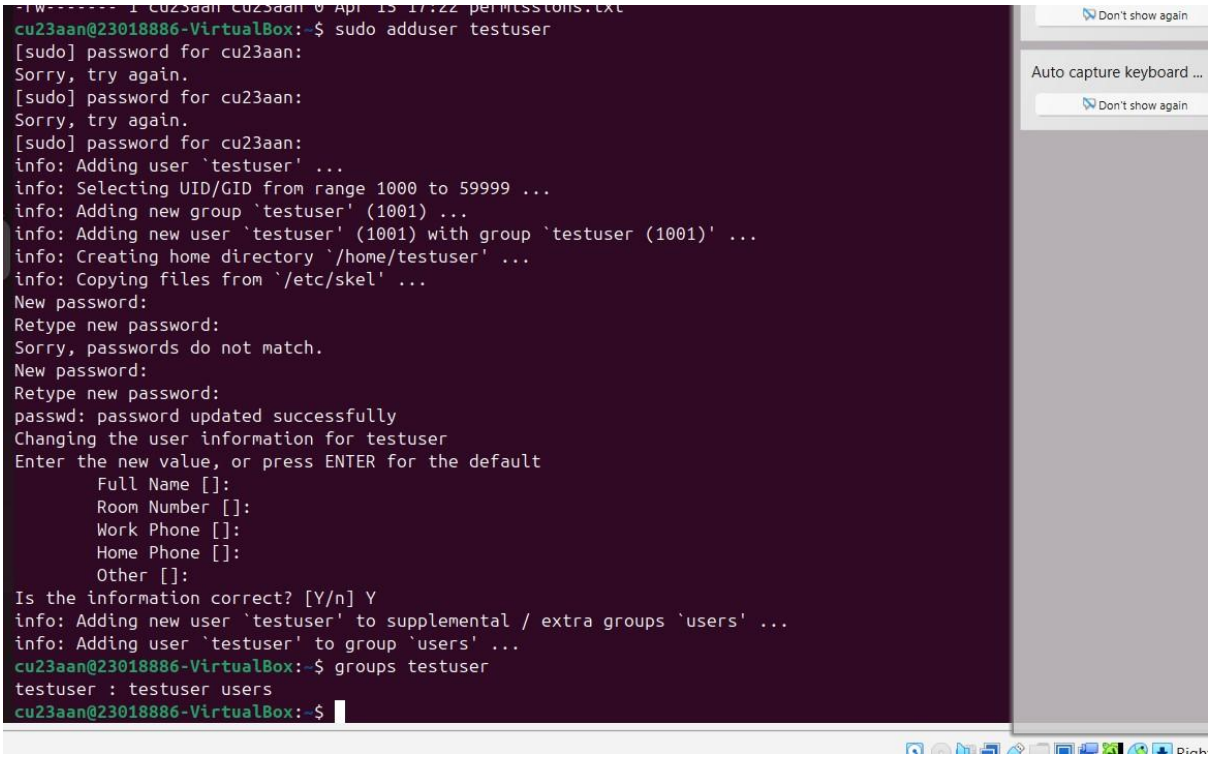


Figure 8: Output showing user creation and group membership

From the output, I could see that testuser was part of the testuser group and the users group. This shows how Linux organises users into groups. Groups are important because permissions can be assigned to a group instead of individual users, which simplifies permission management across multiple users, especially in larger systems.

2.5 File Permissions and How They Protect the System

Linux uses permissions to control who can access files. This includes read (r), write (w), and execute (x) permissions. These permissions are shown when using the ls -l command, which links directly to what I explored earlier.

To demonstrate this, I created a file using: touch permissions.txt

Then I checked its permissions using: ls -l



Figure 9: Output of ls -l before and after changing permissions

From my screenshot, I could see the file had permissions like: -rw-rw-r--

This means: the owner can read and write, group can read and write and others can read only.

I then changed the permissions using: chmod 600 permissions.txt

After running ls -l again, the permissions changed to: -rw-------

This means only the owner can read and write the file, and no one else has access. This showed how the ls -l command and permissions are directly connected, because ls -l is what allows you to see the permission changes after using commands like chmod. This is important for security because it prevents unauthorised users from accessing or modifying files, they should not have access to.

Section 3: DHCP Fundamentals (Concepts + Applications)

3.1 What DHCP is and How it Works?

DHCP is used to automatically assign IP addresses to devices when they connect to a network. Instead of manually entering the IP address, subnet mask, default gateway and DNS, the router does this automatically, reducing the need for manual configuration. The process works in four steps called DORA: Discover (the device broadcasts a request for an IP address), Offer (the server offers an available address), Request (the device requests the offered address), and Acknowledge (the server confirms the assignment). First, the device sends a request because it does not have an IP address. The server then offers one, the device accepts it, and the server confirms it.

Although I did not see each step directly, I could see the result in my PC configuration where the IP address was assigned automatically.

3.2 Why Dynamic IP Allocation is Useful?

DHCP makes network setup easier because everything is assigned automatically. This reduces errors, such as accidentally giving two devices the same IP address, which would stop them from connecting. It is also useful in networks where devices connect and disconnect often, because IP addresses can be reused when devices are no longer connected to the network.

3.3 DHCP Configuration and Evidence

In my Packet Tracer lab, I configured Router0 as the DHCP server for Network A (10.1.1.0/24).

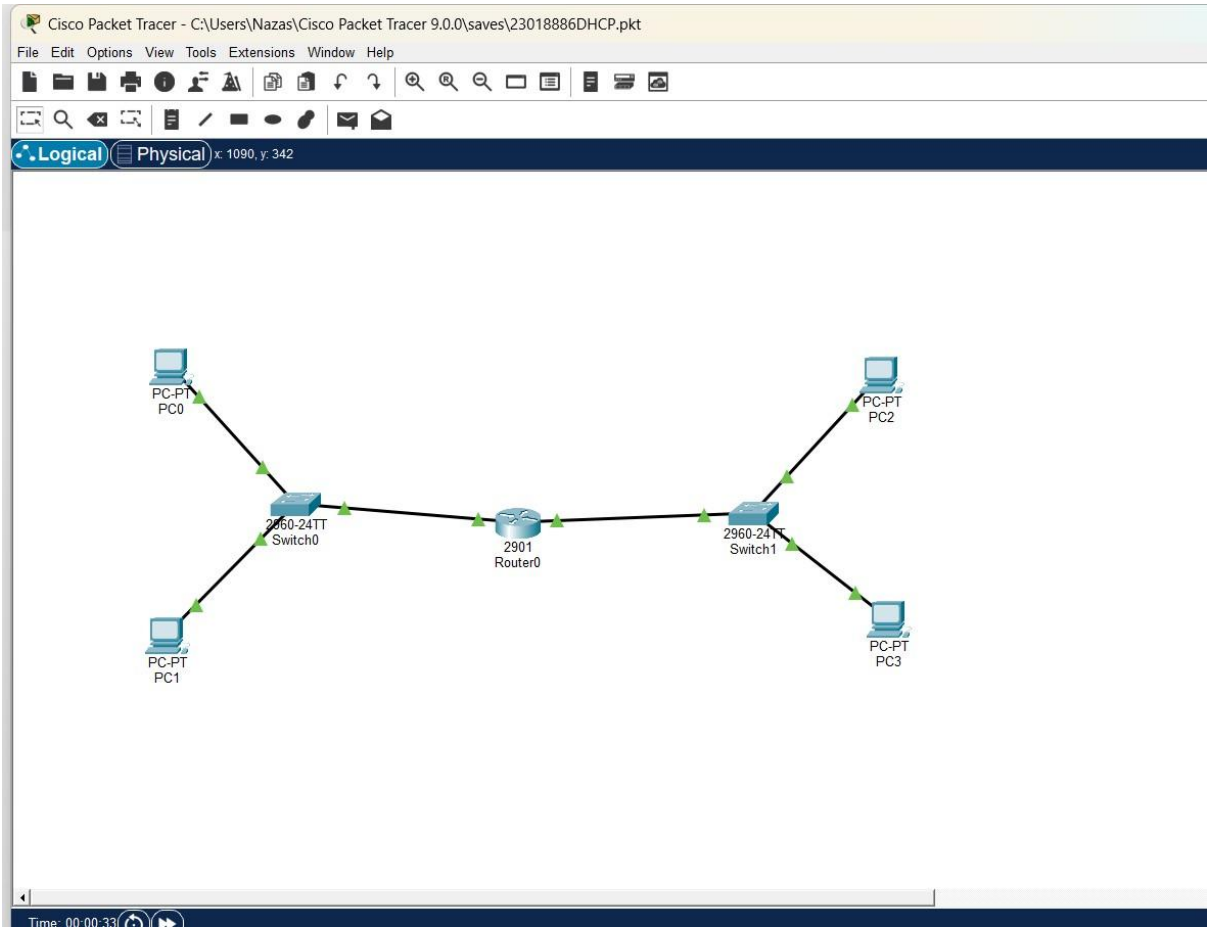


Figure 10: DHCP network topology

I created a DHCP pool called NETWORK_A and set the network range, default gateway, and DNS server. I excluded the router's IP address (10.1.1.1) to avoid conflicts.

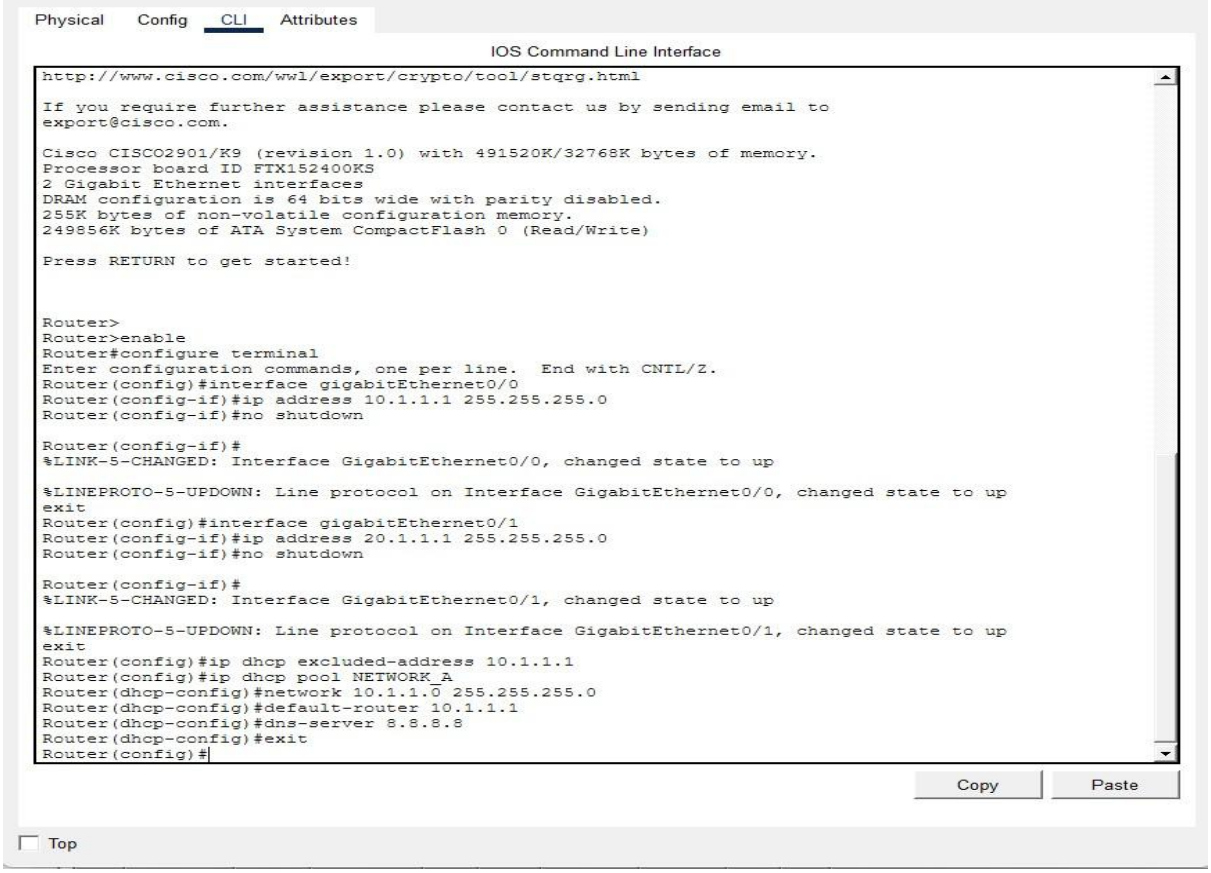


Figure 11: Router CLI showing DHCP configuration

After this, I set PC0 to obtain an IP address automatically.

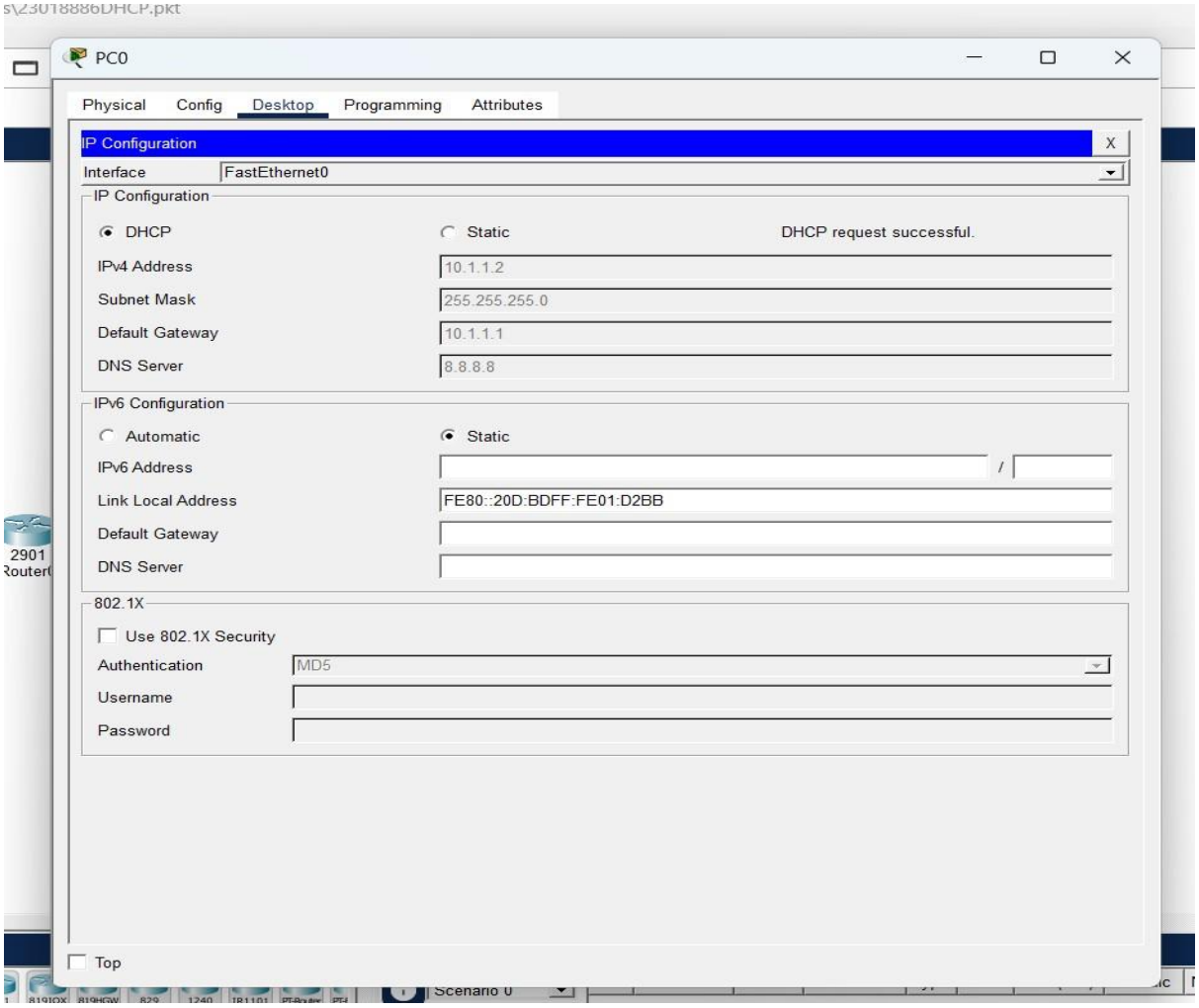


Figure 12: PC0 IP configuration showing DHCP success

The PC received the IP address 10.1.1.2 along with the correct settings, and the message “DHCP request successful” confirmed it worked.

I then used:
show ip dhcp binding

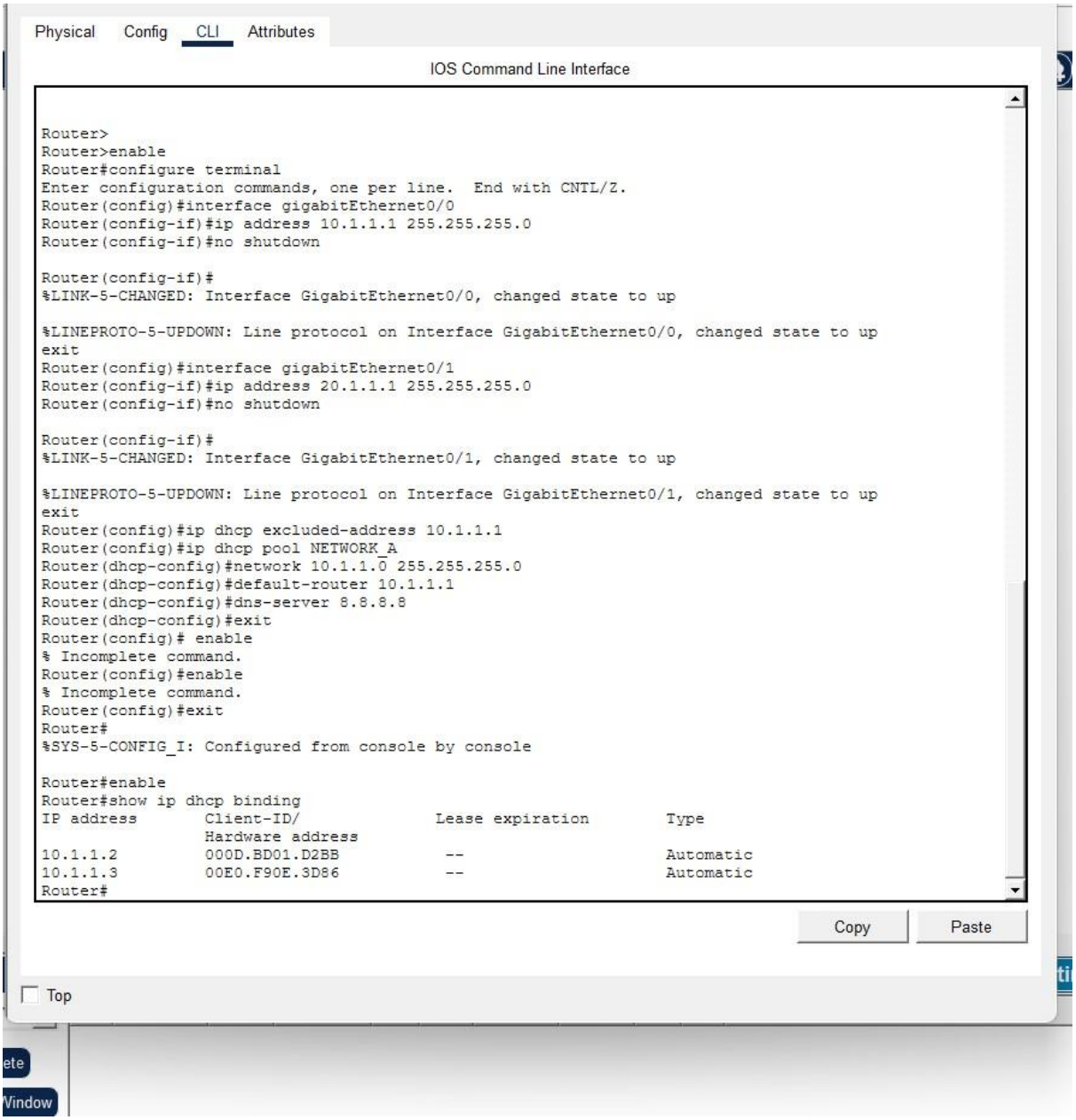


Figure 13: DHCP bindings on the router This showed multiple assigned IP addresses, confirming DHCP was working.

Finally, I tested connectivity using a ping to the default gateway.

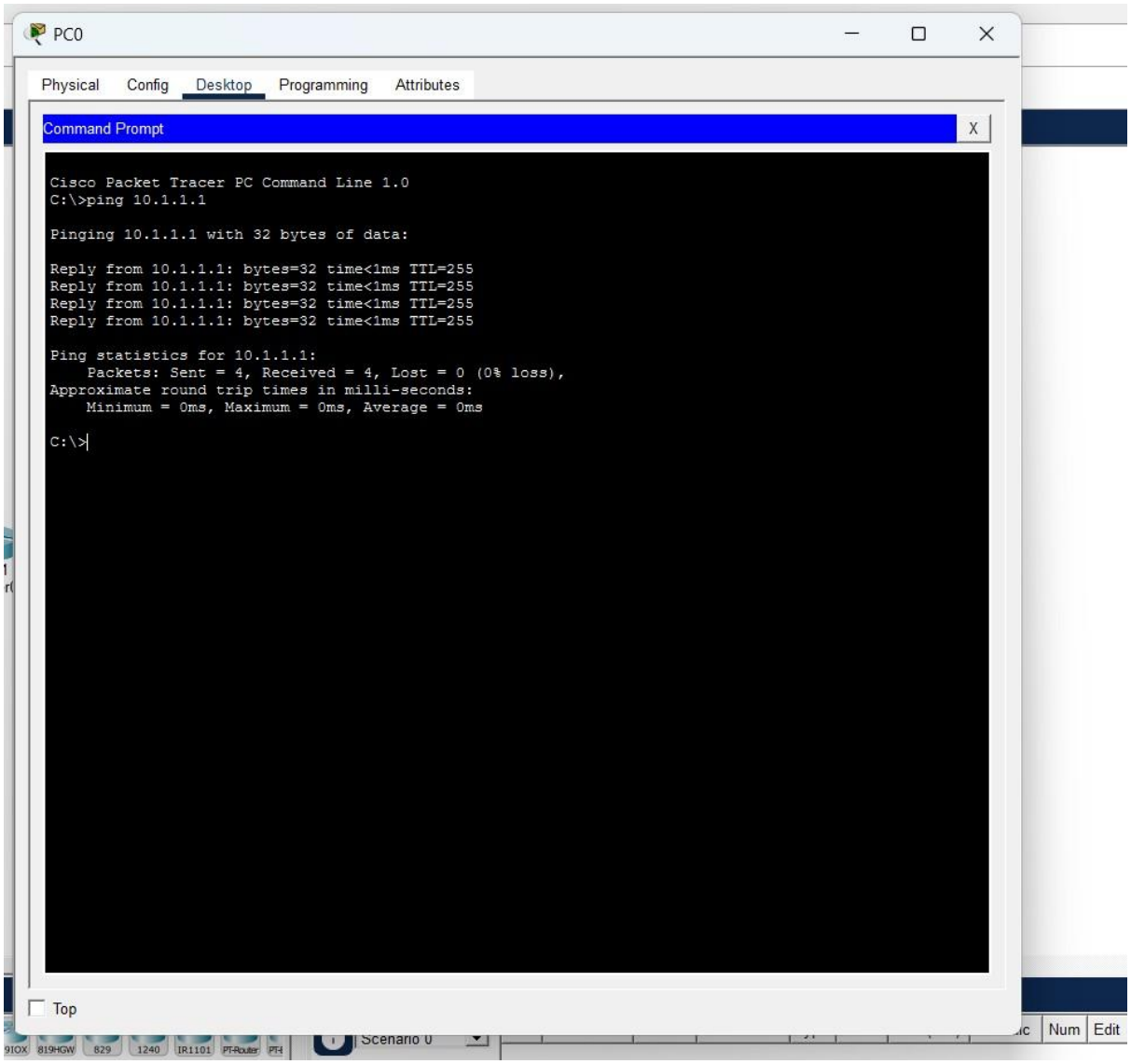


Figure 14: Successful ping to default gateway

The successful replies confirmed the device could communicate properly using the DHCP assigned address.

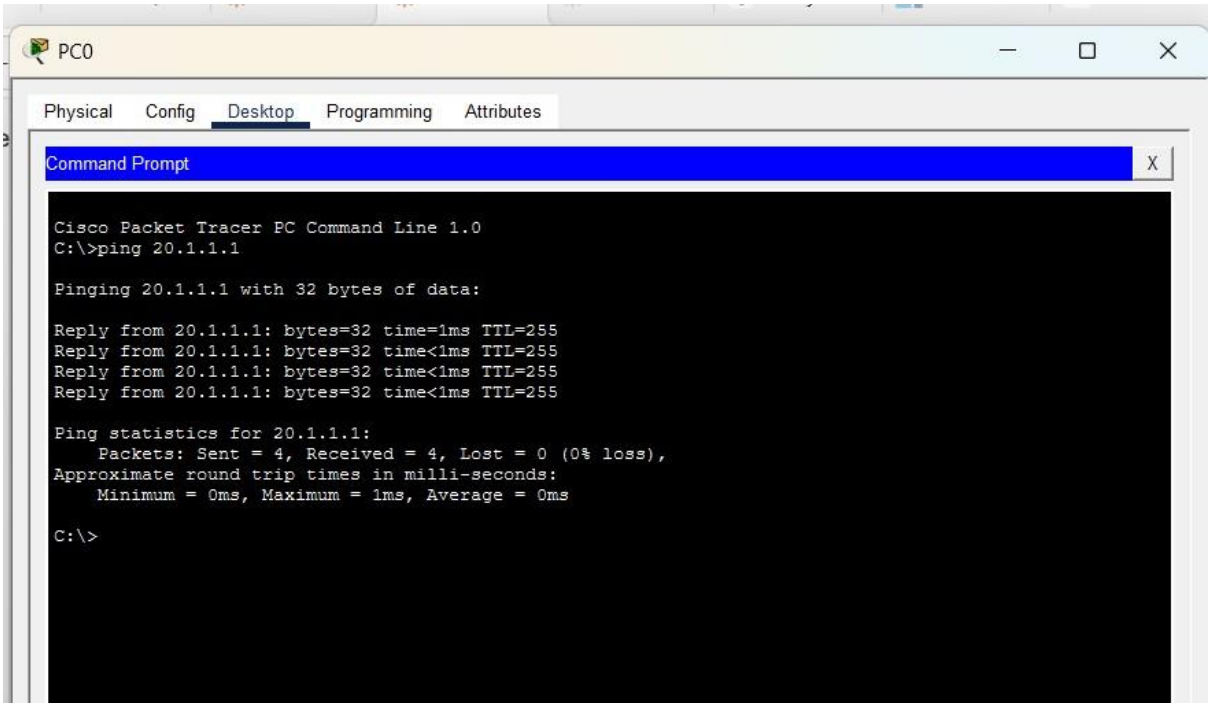


Figure 15: Ping from Network A to Network B confirming cross-network connectivity

From this screenshot, I pinged 20.1.1.1, which is on a different network. The successful replies show that traffic can pass between networks through the router, confirming that DHCP works effectively alongside routing to support communication across networks.

Section 4: Research and Theory Extension

4.1 What a VLAN is?

A VLAN (Virtual Local Area Network) is a way of splitting one physical network into smaller logical networks using the same switch. Even though all the devices are connected to the same switch, they behave as if they are on separate networks.

According to Cisco (2023), VLANs create separate broadcast domains on a single physical switch, which explains why devices are isolated even when connected to the same hardware.

This means devices in one VLAN cannot see or communicate with devices in another VLAN unless routing is added. This is useful because it allows networks to be organised without needing multiple physical switches.

4.2 Why VLANs are useful?

From doing the VLAN lab, I understood that VLANs help organise devices by grouping them logically instead of physically. For example, devices in the ADMIN group can be placed in one VLAN and SALES in another, even if they are connected to the same switch.

VLANs also reduce unnecessary traffic because broadcasts stay within each VLAN instead of being sent to every device. Tanenbaum and Wetherall (2011) explain that this improves performance by limiting broadcast traffic to each VLAN.

Another important thing is security. Devices in different VLANs cannot communicate directly, which helps limit the potential impact of unauthorised access between groups.

4.3 VLAN Lab Implementation

In my Packet Tracer lab, I created two VLANs on a single switch: VLAN 10 named ADMIN and VLAN 20 named SALES. I then assigned the switch ports so that PC0 and PC1 were in VLAN 10, and PC2 and PC3 were in VLAN 20. Cisco (2020) also shows how VLANs are configured on switches using port assignments, which matches what I did in my lab.

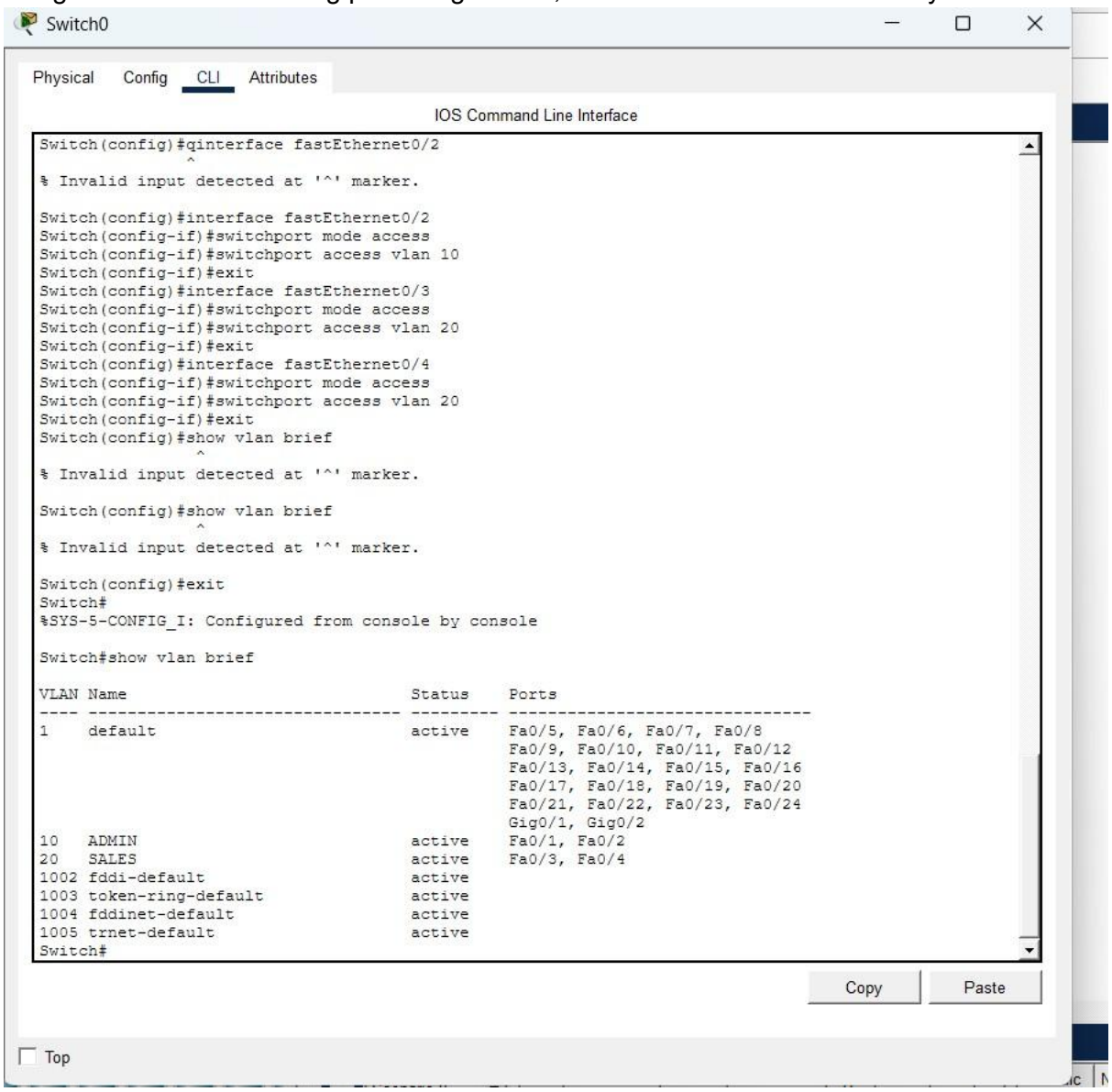


Figure 16: Output of show vlan brief showing VLAN assignments

From my screenshot, I could see that Fa0/1–Fa0/2 were assigned to VLAN 10 and Fa0/3–Fa0/4 were assigned to VLAN 20. This confirmed that the VLAN configuration was correct.

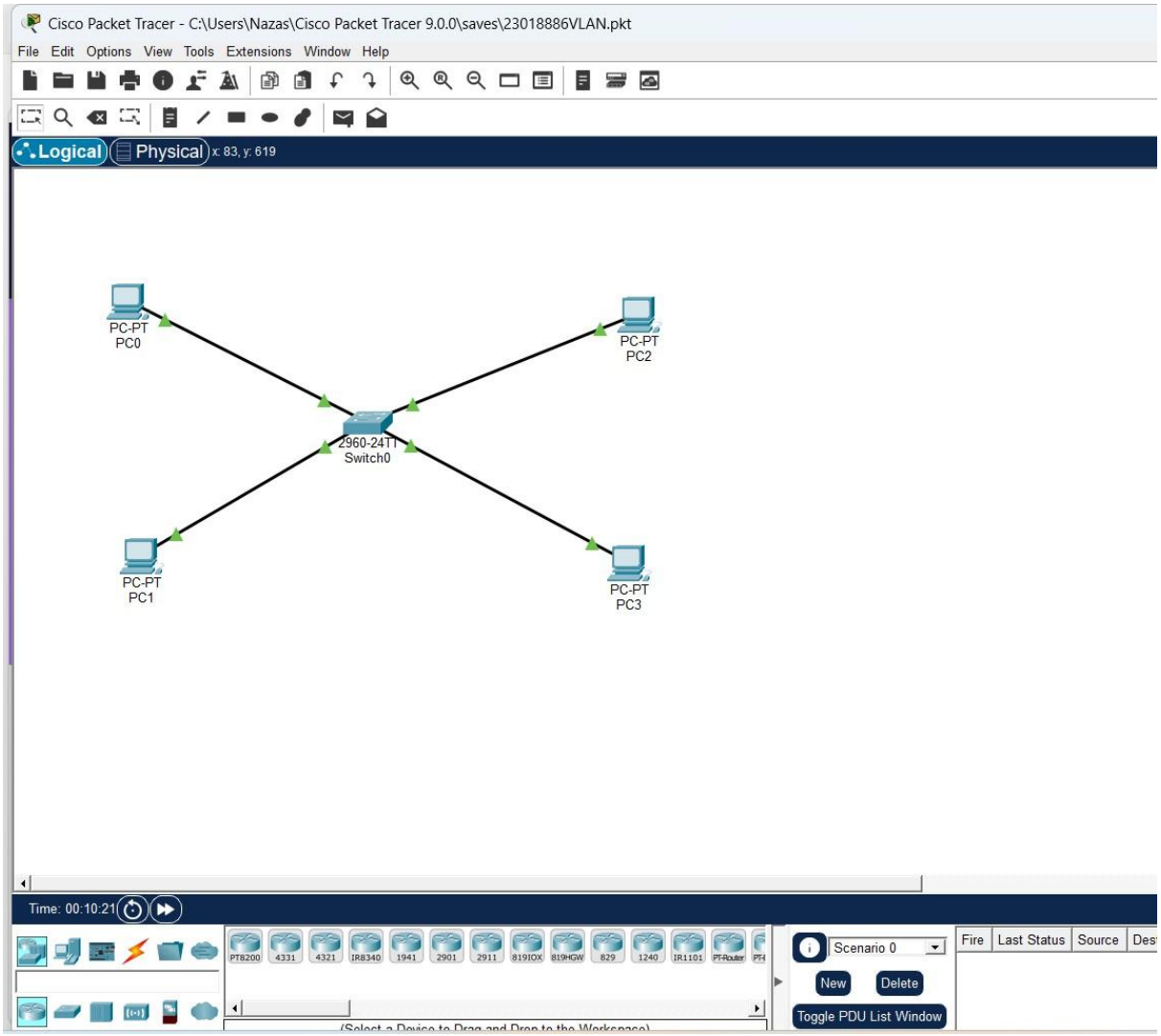


Figure 17: VLAN topology showing switch and connected PCs

This diagram shows how all devices are connected to one switch but separated into different VLANs.

4.4 Testing Communication

To test the VLAN setup, I used ping between devices. First, I tested communication within the same VLAN: PC0 to PC1 (both in VLAN 10)

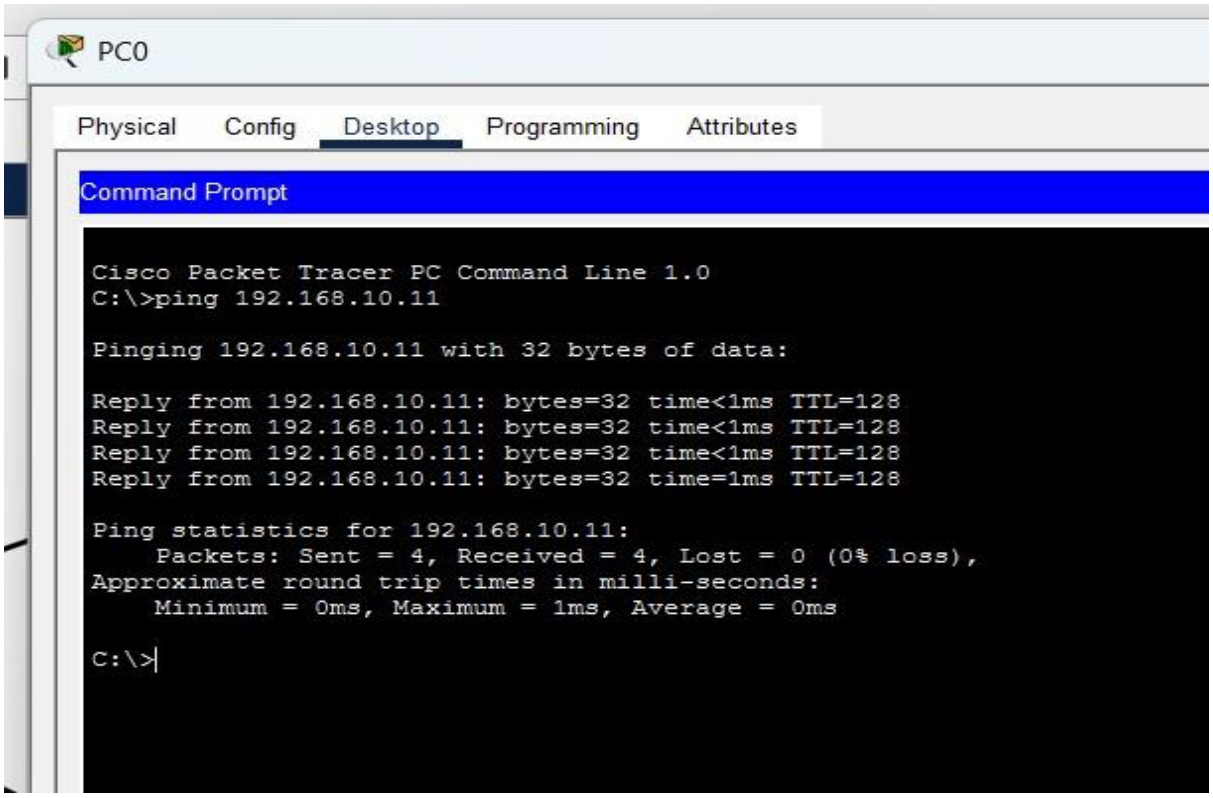


Figure 18: Successful ping within the same VLAN

The ping was successful with 0% packet loss. This shows that devices in the same VLAN can communicate normally without requiring routing. Then I tested communication between different VLANs: PC0 (VLAN 10) to PC2 (VLAN 20)

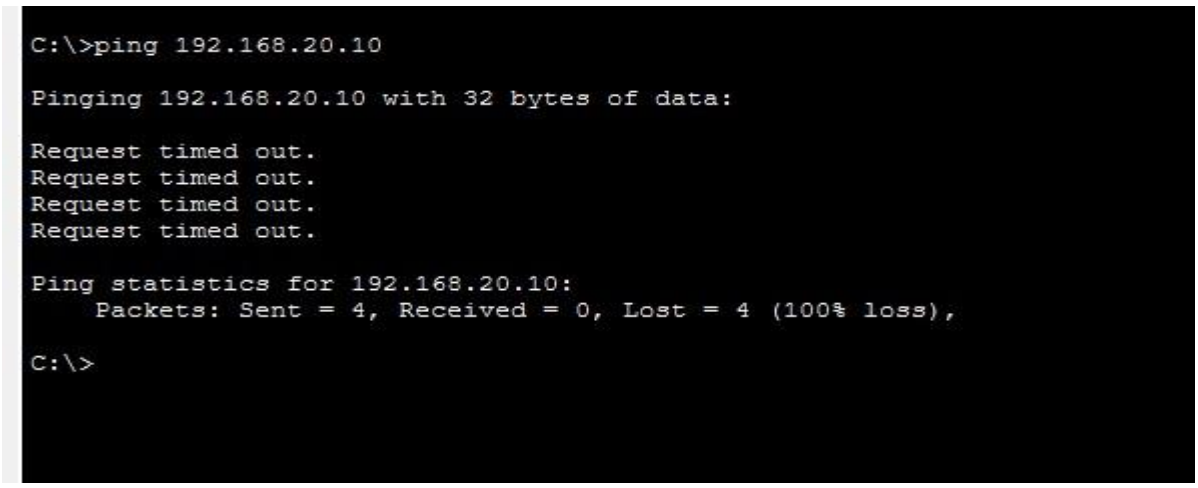


Figure 19: Failed ping between different VLANs

This ping failed completely, which shows that devices in different VLANs cannot communicate without routing. This confirmed that the VLAN segmentation is working correctly. This demonstrates how VLANs enforce network separation in practice, not just in theory.

4.5 Why Inter-VLAN Routing is Needed

From the results, I understood that VLANs separate networks at Layer 2, which is why devices in different VLANs cannot communicate.

According to Lammle (2016), inter-VLAN routing requires a Layer 3 device, such as a router, to forward traffic between VLANs. This matches what I observed in my lab, where communication between VLANs failed without routing. This is called inter-VLAN routing. Without it, each VLAN remains separate, which can limit communication and make larger networks harder to manage.

4.6 Limitations of VLANs

One limitation I noticed is that VLANs must be configured correctly. If ports are assigned to the wrong VLAN, devices may not communicate properly.

They can also become more complex as the network grows, especially with multiple switches where VLAN configuration must be consistent across all devices. This shows that while VLANs improve security and organisation, they also introduce additional configuration complexity when communication between networks is required.

Conclusion

This report has explained key networking and operating system concepts based on the practical work I completed during the module. In Section 1, I showed how a routed network was set up in Packet Tracer and how routing using RIP allowed devices in different networks to communicate, supported by successful ping results.

In Section 2, I used the Linux terminal to explore the file system, users, and permissions. The screenshots showed how commands such as ls -l, whoami, and chmod were used to manage files and control access.

In Section 3, I configured DHCP on a router and tested it by setting a PC to automatically receive an IP address. The results matched the configuration and were confirmed using both the IP settings and successful ping tests.

In Section 4, I created VLANs and tested communication between devices. The results showed that devices in the same VLAN could communicate, while devices in different VLANs could not, unless routing is used.

Overall, the practical work helped me understand how networks are configured, tested, and managed, and how different concepts such as routing, DHCP, and VLANs work together in real systems.

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